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1: INTRODUCTION

Task management is an essential part of both personal and professional life. Students, employees, and organizations often work on multiple tasks simultaneously, making it challenging to organize activities, prioritize important work, and meet deadlines. Traditional methods such as paper notes, notebooks, spreadsheets, and basic to-do lists are often difficult to maintain, especially when the number of tasks increases. These methods lack flexibility, real-time organization, and an efficient way to track the progress of ongoing work.

With the advancement of web technologies, modern task management systems have become more interactive, user-friendly, and accessible. Most of these applications follow the **Kanban methodology**, a visual workflow management technique that organizes tasks into different stages such as **To Do**, **Doing**, and **Done**. This approach allows users to monitor task progress easily, improve productivity, and manage workloads in a structured manner.

The **Trello Clone – Task Management System** is a web-based application developed to simplify task organization and project management. The system provides an intuitive interface where users can create multiple boards for different projects, organize work into lists, and manage tasks using interactive task cards. Each task can include a title, description, priority level, and due date, enabling users to maintain detailed information about their work.

To enhance usability, the application includes features such as **drag-and-drop functionality**, allowing users to move tasks between different stages of completion with ease. A built-in **search feature** helps users quickly locate tasks, while **Light Mode and Dark Mode** provide a comfortable viewing experience under different lighting conditions. The application also uses the browser's **Local Storage** to save user data, ensuring that boards, lists, and tasks remain available even after refreshing or reopening the browser.

The project has been developed using **HTML5**, **CSS3**, and **JavaScript**, following a modular structure that improves readability, maintainability, and future scalability. The responsive design ensures that the application works efficiently across desktops, laptops, tablets, and mobile devices.

Overall, the **Trello Clone – Task Management System** provides a simple, efficient, and modern solution for organizing daily activities, academic assignments, and small projects. It demonstrates the practical implementation of modern web development technologies while offering a user-friendly platform that improves productivity and task management.

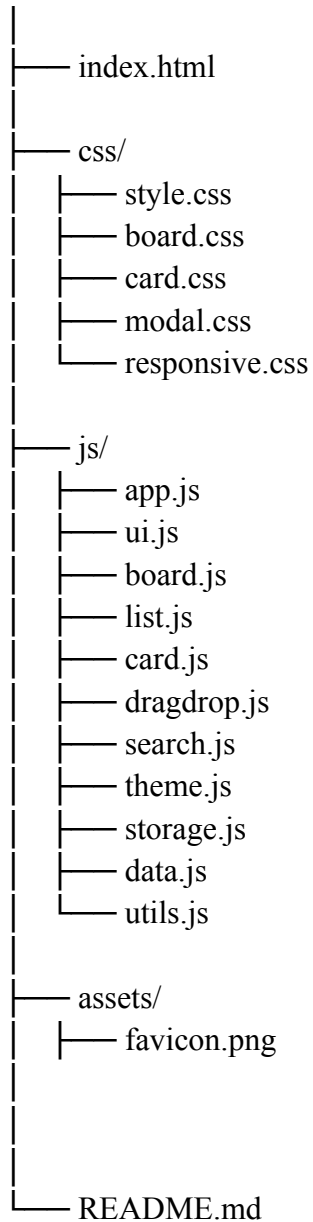
2. TECHNOLOGIES USED

Technologies	Purpose
HTML5	Used to create the structure of the web pages.
CSS3	Used to design the user interface and make it responsive
JavaScript	Used to add functionality like creating boards, lists, tasks, search, drag & drop, and dark mode.
Local Storage	Used to save data in the browser without a database.
Visual Studio Code	Used to write and manage the project code.

3. PROJECT STRUCTURE

The Trello clone project follows a modular folder structure to improve code organization, readability, and maintainability. Each folder contains files responsible for specific functionalities of the application.

TMS/



4. IMPLEMENTATION

4.1 Overview

The **Trello Clone – Task Management System** is implemented as a web-based application using HTML5, CSS3, and JavaScript. The application follows a modular structure where each JavaScript file is responsible for a specific functionality. User data is stored using the browser's Local Storage, allowing tasks to persist even after the page is refreshed.

4.2 User Interface Implementation

The user interface is developed using HTML5 and CSS3 to provide a clean, responsive, and user-friendly design. The application consists of a sidebar for board management, a header containing the board title, search bar, and theme toggle button, and a board area displaying task lists and cards.

Features Implemented:

- Responsive layout
 - Sidebar navigation
 - Modern task board interface
 - Modal windows for creating boards, lists, and tasks
 - Light and Dark mode
-

4.3 Board Management

The Board Management module allows users to organize different projects by creating multiple boards.

Functions Implemented:

- Create Board
- Rename Board
- Delete Board
- Switch between Boards

4.4 List Management

Each board contains multiple task lists such as **To Do**, **Doing**, and **Done**.

Functions Implemented:

- Create List
 - Rename List
 - Delete List
-

4.5 Task (Card) Management

Each list contains task cards that represent individual activities.

Functions Implemented:

- Add Task
 - Edit Task
 - Delete Task
 - Set Task Priority
 - Add Due Date
 - Add Task Description
-

4.6 Drag and Drop

Drag-and-drop functionality is implemented using JavaScript Drag Events. Users can move task cards between different lists by dragging and dropping them. The updated task position is automatically saved in Local Storage.

4.7 Search Functionality

A search feature is implemented to quickly locate tasks. Users can search by entering keywords, and matching task cards are displayed instantly.

4.8 Theme Management

The application provides both Light Mode and Dark Mode. Users can switch themes using the theme button, and the selected theme is saved using Local Storage.

4.9 Local Storage Implementation

The application uses the Local Storage API to save all project data.

The following information is stored:

- Boards
- Lists
- Tasks
- User-selected theme

This allows the application to restore data automatically whenever the user reopens the application.

4.10 Project Output

The implemented system successfully provides:

- Multiple Board Management
- List Management
- Task Management
- Drag and Drop
- Search Tasks
- Light/Dark Theme

- Responsive User Interface
- Local Storage Support

The application is lightweight, easy to use, and suitable for students and individuals to organize daily tasks and small projects efficiently.

5. FEATURES OF THE SYSTEM

The **Trello Clone – Task Management System** is designed to help users organize and manage their daily tasks efficiently. The system provides a simple, responsive, and user-friendly interface with the following features:

5.1 Board Management

- Create multiple boards for different projects.
- Rename existing boards.
- Delete boards when they are no longer required.
- Switch between different boards easily.

5.2 List Management

- Create multiple task lists within each board.
- Organize tasks into **To Do**, **Doing**, and **Done** lists.
- Rename task lists.
- Delete task lists.

5.3 Task (Card) Management

- Add new tasks.
- Edit task details.
- Delete completed or unwanted tasks.
- Add task descriptions.
- Set task priorities (Low, Medium, High).
- Assign due dates to tasks.

5.4 Drag and Drop

- Move tasks between different lists using drag-and-drop.
- Update task progress quickly and easily.
- Automatically save changes after moving tasks.

5.5 Search Functionality

- Search tasks by title or description.
- Display matching tasks instantly.

- Makes it easy to locate tasks in large boards.

5.6 Dark Mode

- Switch between Light Mode and Dark Mode.
- Improves readability in different lighting conditions.
- Saves the selected theme using Local Storage.

5.7 Local Storage Support

- Stores all boards, lists, and tasks in the browser.
- Preserves user data after refreshing or reopening the application.
- Does not require a database or internet connection.

5.8 Responsive Design

- Supports desktop, laptop, tablet, and mobile devices.
- Automatically adjusts the layout based on screen size.
- Provides a consistent user experience across devices.

5.9 User-Friendly Interface

- Simple and intuitive navigation.
- Modern and clean design.
- Easy-to-use buttons, icons, and menus.
- Interactive interface for efficient task management.

Summary of Features

- Multiple Board Management
- List Management
- Task CRUD (Create, Read, Update, Delete)
- Drag-and-Drop Functionality
- Task Search
- Priority and Due Date Management
- Light and Dark Theme
- Local Storage Support
- Responsive User Interface
- Modern Dashboard Design

6. RESULTS AND SCREENSHOTS

The **Trello Clone – Task Management System** was successfully designed and implemented using HTML5, CSS3, and JavaScript. The application provides an intuitive and responsive interface for managing daily tasks and projects. All the planned features were implemented successfully and tested under different user interactions.

The application allows users to create multiple boards, organize tasks into different lists, add and manage task cards, search tasks, switch between Light and Dark modes, and move tasks between lists using drag-and-drop functionality. User data is stored in the browser using Local Storage, ensuring that information remains available even after refreshing or reopening the application.

Results Obtained

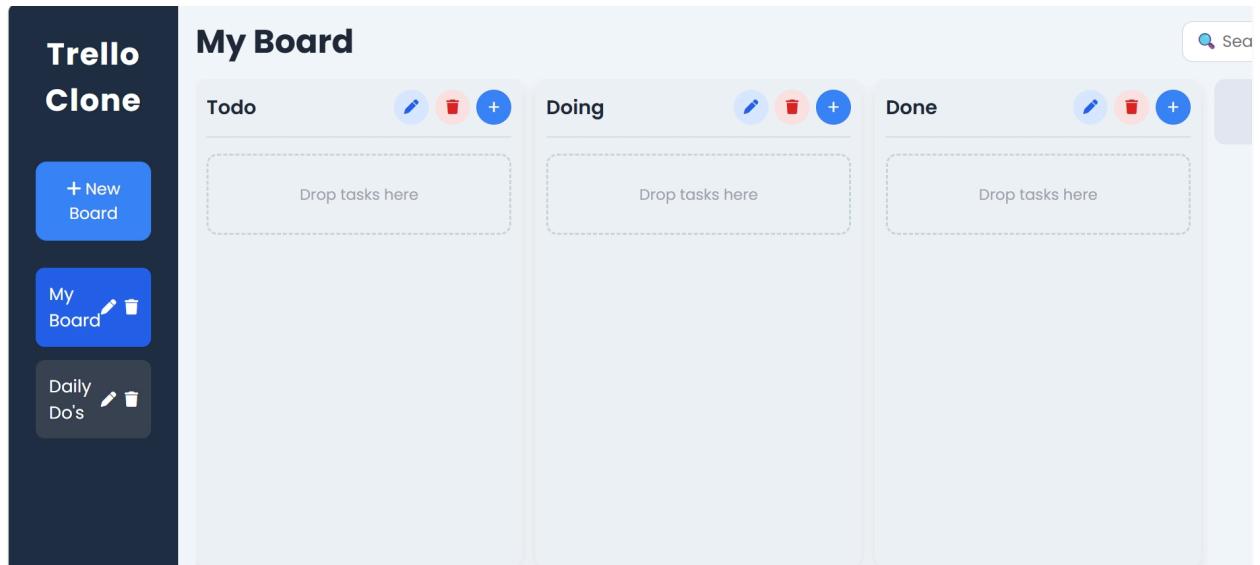
- Successfully created multiple project boards.
- Successfully created, renamed, and deleted task lists.
- Successfully added, edited, and deleted task cards.
- Successfully implemented drag-and-drop functionality for moving tasks.
- Successfully implemented task search functionality.
- Successfully implemented Light Mode and Dark Mode.
- Successfully stored and retrieved application data using Local Storage.
- Successfully developed a responsive interface for desktop and mobile devices.

Performance

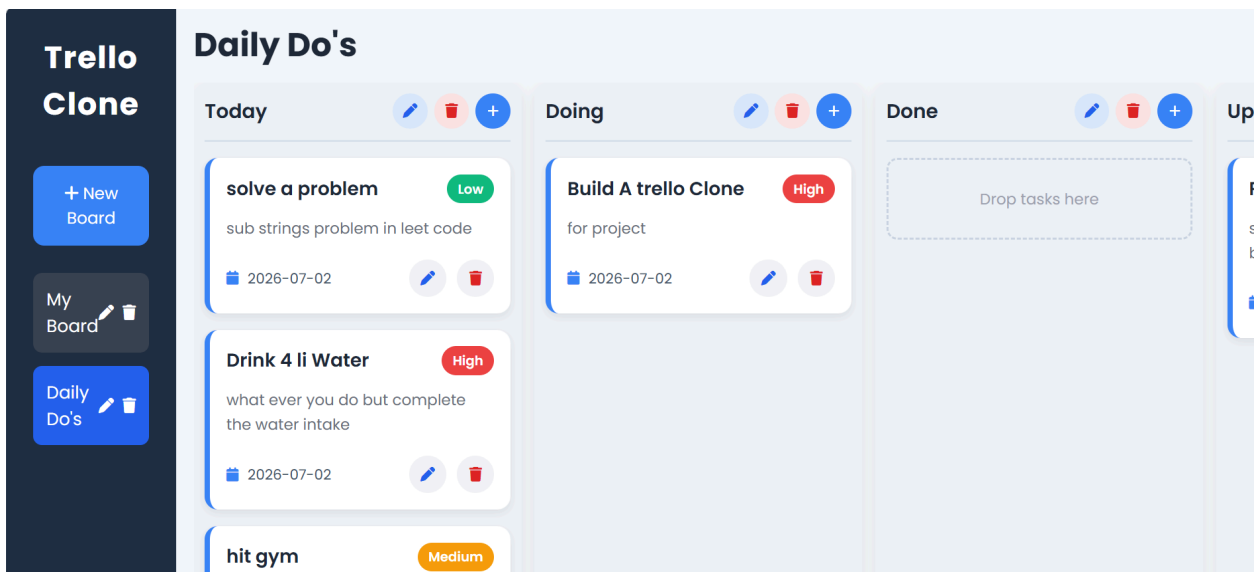
The application performs efficiently because all operations are executed on the client side without requiring a backend server. Task updates, drag-and-drop operations, and search functionality respond instantly, providing a smooth user experience.

Output ScreenShots

The following screenshots demonstrate the successful implementation of the TaskFlow application:



Home Page



After Adding Task's

7. ADVANTAGES

The **TaskFlow – Task Management System** offers several advantages that make task organization simple, efficient, and user-friendly. The major advantages of the system are listed below:

- **Easy Task Management:** Users can create, edit, delete, and organize tasks quickly and efficiently.
- **Multiple Board Support:** Different boards can be created to manage multiple projects or categories separately.
- **Organized Workflow:** Tasks are arranged into **To Do**, **Doing**, and **Done** lists, making it easy to track progress.
- **Drag-and-Drop Functionality:** Tasks can be moved between lists using drag-and-drop, providing a simple and interactive user experience.
- **Quick Search:** Users can search tasks instantly by entering keywords, saving time when managing a large number of tasks.
- **Responsive Design:** The application works smoothly on desktops, laptops, tablets, and mobile devices.

- **Dark Mode Support:** Users can switch between Light Mode and Dark Mode for a comfortable viewing experience.
- **Local Storage Support:** All boards, lists, and tasks are stored in the browser, allowing data to remain available even after refreshing the page.
- **Simple and User-Friendly Interface:** The clean and modern interface makes the application easy to learn and use.
- **Fast Performance:** Since the application runs entirely on the client side, task operations are performed quickly without requiring a backend server.
- **Low Cost:** The application is developed using free and open-source web technologies, making it economical to build and maintain.
- **Easy Maintenance:** The project follows a modular folder structure, making future updates and feature additions easier.
- **Improved Productivity:** Users can organize daily activities, monitor task progress, and complete work more efficiently.

8. LIMITATIONS

Although the **TaskFlow – Task Management System** provides an efficient solution for task management, the current version has the following limitations:

1. **No User Authentication:** The application does not provide user registration or login functionality.
2. **No Cloud Storage:** All data is stored in the browser using Local Storage and cannot be accessed from other devices.
3. **Single User System:** The application is designed for individual users and does not support multiple users or team collaboration.
4. **No Real-Time Synchronization:** Changes made by one user cannot be viewed by others in real time.
5. **Limited Task Features:** The system does not support file attachments, comments, labels, or checklists for tasks.
6. **No Notification System:** The application does not provide reminders or notifications for upcoming task deadlines.

7. **No Data Backup or Export:** Users cannot export their task data to PDF, Excel, or other file formats, and there is no automatic backup feature.
8. **Internet-Based Browser Application:** The project is designed to run in a web browser and does not include a dedicated mobile or desktop application.

Summary

These limitations do not affect the core functionality of the application but provide opportunities for future improvements. Features such as user authentication, cloud storage, team collaboration, notifications, and data export can be added in future versions to enhance the overall functionality and user experience.

9: FUTURE ENHANCEMENTS

The current version of the **Trello Clone – Task Management System** provides the essential features required for efficient task management. However, the application can be further improved by implementing the following enhancements:

1. **User Authentication**
 - Add user registration and login functionality to provide secure access to personal task boards.
2. **Cloud Database Integration**
 - Store data in cloud databases such as Firebase or MongoDB so users can access their tasks from any device.
3. **Real-Time Collaboration**
 - Allow multiple users to work on the same board simultaneously with real-time updates.

4. **Task Notifications and Reminders**

- Send notifications or email reminders for upcoming task deadlines and important activities.

5. **File Attachments**

- Allow users to upload and attach documents, images, or other files to task cards.

6. **Labels and Categories**

- Add colored labels, tags, and categories to improve task organization and filtering.

7. **Calendar Integration**

- Display tasks in a calendar view to help users manage schedules and deadlines more effectively.

8. **Data Export and Backup**

- Provide options to export task data to PDF or Excel and create automatic backups.

9. **Mobile Application**

- Develop Android and iOS versions of the application for improved accessibility and convenience.

10. **Analytics Dashboard**

- Include charts and statistics showing completed tasks, pending tasks, productivity trends, and project progress.

Summary

These future enhancements will improve the functionality, scalability, and usability of the TaskFlow – Task Management System. Implementing these features will transform the application from a personal task management tool into a comprehensive project management platform suitable for individuals, teams, and organizations.

10. CONCLUSION

The **Trello Clone – Task Management System** was successfully designed and developed as a web-based application to simplify task and project management. The project provides an efficient and user-friendly platform for creating boards, organizing tasks into lists, managing task details, and tracking progress using a Kanban-style workflow.

The application includes essential features such as **Board Management, List Management, Task (Card) Management, Drag-and-Drop functionality, Task Search, Light and Dark Mode, Responsive Design**, and **Local Storage** for data persistence. These features enable users to organize their daily activities effectively while maintaining a simple and intuitive user interface.

The project demonstrates the practical application of modern web technologies including **HTML5**, **CSS3**, and **JavaScript** to build an interactive and responsive application without requiring a backend server. The modular project structure also makes the application easy to maintain, understand, and enhance in the future.

Although the current system has certain limitations, it provides a strong foundation for future improvements such as user authentication, cloud storage, real-time collaboration, notifications, file attachments, and mobile application support.

In conclusion, the **TaskFlow – Task Management System** successfully achieves its objectives by providing a reliable, responsive, and easy-to-use task management solution. The project enhances productivity, improves task organization, and demonstrates the effective use of modern web development technologies in solving real-world task management problems.

11.REFERENCES

The following resources were referred to during the design and development of the **Trello Clone – Task Management System**:

1. Mozilla Developer Network (MDN) Web Docs – HTML, CSS, and JavaScript Documentation.
2. W3Schools – HTML, CSS, JavaScript, and Local Storage Tutorials.
3. ECMAScript (JavaScript) Documentation.
4. Font Awesome Documentation – Icons and UI Components.
5. Google Fonts Documentation – Typography and Web Fonts.
6. Visual Studio Code Documentation.

7. GitHub Documentation – Version Control and Repository Management.
8. HTML5 Official Specification – World Wide Web Consortium (W3C).
9. CSS3 Official Specification – World Wide Web Consortium (W3C).
10. ChatGPT (OpenAI) – Used for guidance in project planning, debugging, documentation preparation, and code explanation.